

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTESC305 - DIGITAL ELECTRONICS AND MICROPROCESSOR

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The Gray code of binary 1011 is:
a) 1111 b) 1110 c) 1001 d) 1010
- 2 What is the decimal equivalent of hexadecimal number 1F?
a) 15 b) 31 c) 25 d) 35
- 3 Which of the following represents the standard SOP form?
a) $A + BC$ b) $AB + A'C$ c) $A + B$ d) $AB(C + D)$
- 4 Which IC is commonly used for parity generation and checking?
a) 7486 b) 74154 c) 74151 d) 74138
- 5 A 1-bit memory cell can store:
a) 0 only b) 1 only c) Either 0 or 1 d) Multiple bits
- 6 In a Mealy FSM, the output depends on:
a) State only b) Input only
c) Present input and current state d) Next state only
- 7 The flag register in 8085 consists of how many flags?
a) 3 b) 5 c) 6 d) 8
- 8 The opcode fetch (OF) machine cycle requires how many T-states?
a) 2 b) 3 c) 4 d) 6
- 9 Which hardware interrupt in 8085 has the highest priority?
a) INTR b) RST 6.5 c) RST 5.5 d) TRAP
- 10 The 8085 microprocessor has how many hardware interrupts?
a) 1 b) 3 c) 5 d) 8
- 11 How many ports does the 8255 provide for I/O operations?
a) 1 b) 2 c) 3 d) 4
- 12 The stack in 8085 grows in which direction?
a) Upward b) Downward c) Randomly d) Depends on program

Q2. Solve the following.

- A) Convert the Boolean function $F(A, B, C) = A'B + AB'C + ABC$ into its minimized form using Boolean laws. 6
- B) Simplify the function $F(A, B, C, D) = \Sigma(1, 3, 7, 11, 15)$ using a 4-variable K-map. 6

Q3. Solve the following.

- A) Design a 4-to-1 multiplexer using basic gates. 6
- B) Design a 2-bit comparator using logic gates. Explain the logic for equal, greater, and lesser conditions. 6

Q4. Solve Any Two of the following.

- A) Design a 4-bit SIPO (Serial-In Parallel-Out) shift register using D flip-flops. 6
- B) Design a Moore FSM to detect the sequence 101 in a serial input stream. 6
- C) Discuss the different types of shift registers. 6

Q5. Solve Any Two of the following.

- A) Describe following instructions of 8085 microprocessor with examples of each. 6
 - a) LDAX b) LHLD c) LXI
- B) Draw timing diagram of LHLD instruction. 6
- C) Describe functions of following pins of 8085 microprocessor. 6
 - ALE, IO/M, INTA, SID, HOLD & Ready

Q6. Solve Any Two of the following.

- A) Explain the process of serial data transfer and reception in the 8085 microprocessor with example. 6
- B) Write an 8085 assembly language program to multiply content of B register with C register. 6
- C) Compare Memory Mapped I/O and I/O Mapped I/O with examples. 6

*** End ***